

1 Gaussian Plume Model

Atmospheric dispersion from point sources

1.1 Wind direction and variability

Prediction. Predict the plan-view plume shape for constant northerly wind and for a prevailing northerly wind with directional variability.

Expected prediction. A constant wind should produce a narrow, straight plume aligned with the mean downwind direction. A prevailing wind with directional variability should produce a broader, fan-shaped time-mean footprint because individual hourly plumes point in slightly different directions.

Exercise 1. How does wind-direction variability change the area exposed to pollutant and the maximum time-mean concentration?

Model answer. The constant-wind case produces a narrow high-concentration plume directly downwind. With a prevailing but variable wind, the pollutant is spread over a much larger area and the time-mean peak concentration is lower because the plume does not pass over the same line on every timestep. The underlying hourly plume is still Gaussian; the broader footprint arises from averaging plumes with different wind directions.

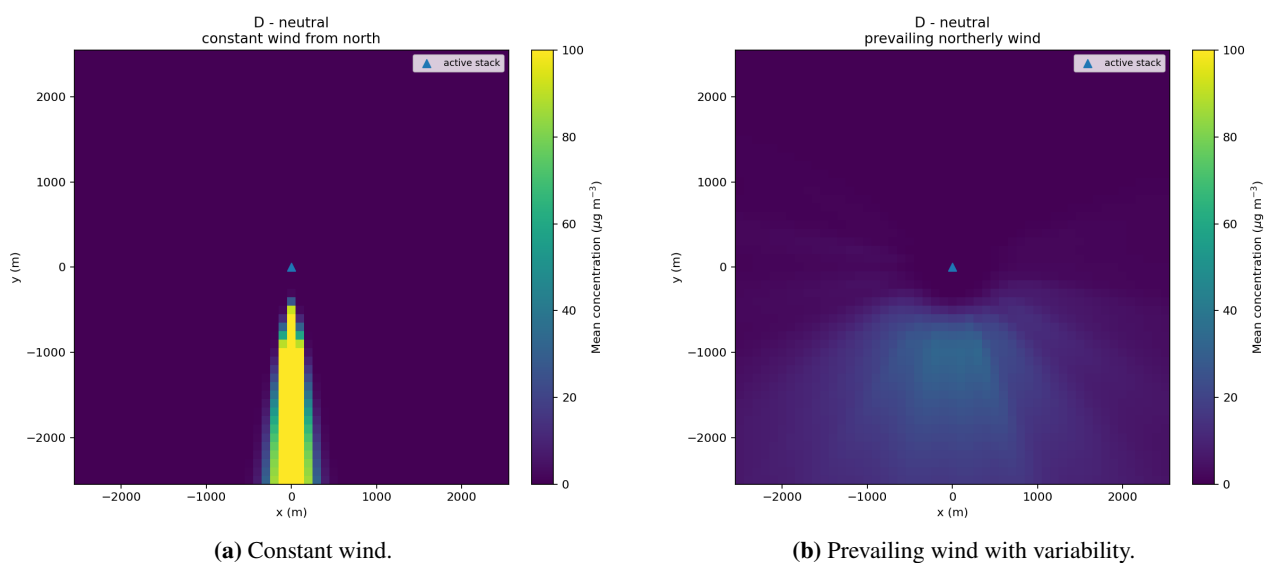


Figure 1: Directional variability broadens the time-mean footprint and reduces the maximum concentration.

1.2 Multiple stacks

Exercise 2. Where do the separate plume fields overlap, and why can their concentrations simply be added?

Model answer. The three plumes overlap mainly in the common downwind sector, so concentrations are enhanced where two or more plume footprints coincide. The Gaussian plume equation used here is linear in concentration and in source strength. There are no chemical reactions, saturation effects or interactions between plumes, so if C_1 , C_2 and C_3 are solutions for the individual sources, $C_1 + C_2 + C_3$ is also the solution for the combined sources.

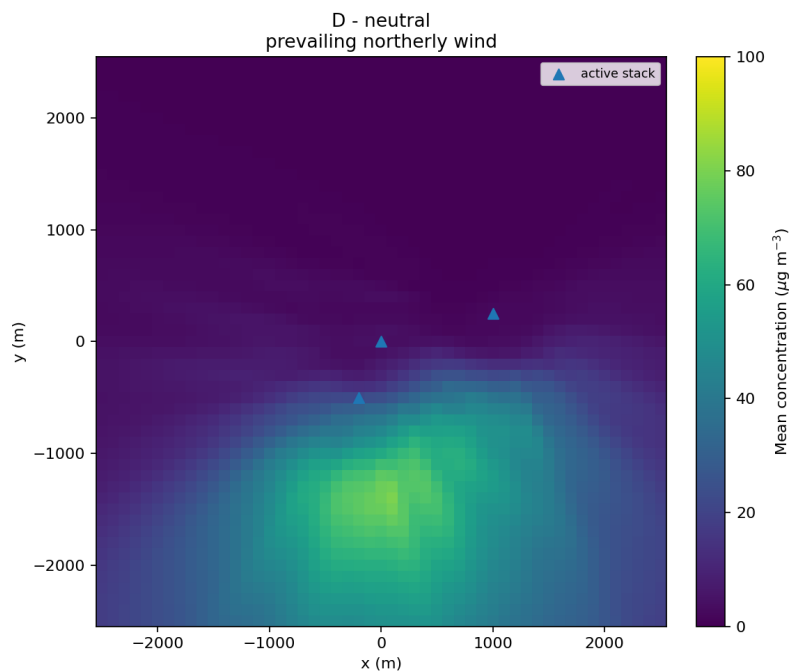


Figure 2: Three-stack prevailing-wind calculation. The fields overlap in the common downwind region and add linearly.

1.3 Atmospheric stability: surface plan view

Exercise 3. Compare stability classes A, D and F. How do plume width, downwind extent and ground-level concentration change, and is more instability always associated with higher concentration at a receptor?

Model answer. Class A (very unstable) has the largest σ_y and σ_z : the plume spreads rapidly both laterally and vertically, producing a broad footprint close to the source but strong dilution with distance. Class D is intermediate. Class F (very stable) has much smaller dispersion parameters, so the plume remains narrow and can persist farther downwind along its centreline. Therefore “more unstable” does not mean “higher concentration everywhere”. Strong mixing can increase the chance that pollutant reaches some nearby ground locations while simultaneously lowering concentration at a particular receptor by spreading the same emitted mass through a much larger volume.

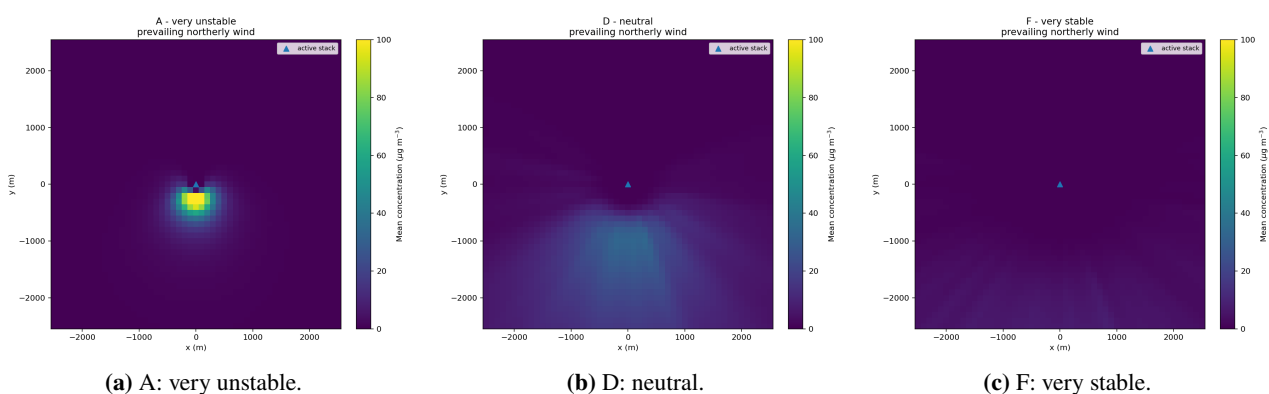


Figure 3: The same emission produces very different ground-level footprints as atmospheric stability changes.

1.4 Atmospheric stability: vertical structure

Exercise 4. How does stability alter the vertical spread of the plume and the fraction of pollutant that reaches the surface? Relate the result to looping, coning and fanning.

Model answer. The class-A plume spreads through a deep vertical layer and reaches the ground strongly near the source because vigorous turbulent mixing rapidly transports pollutant both upward and downward. This corresponds qualitatively to the broad vertical excursions associated with an unstable/“looping” regime, although the model shows only a steady time mean. A neutral atmosphere would give an intermediate, roughly “coning” plume. In class F, vertical spread is weak and the plume forms a thin, horizontally extended layer, qualitatively resembling “fanning”. The small σ_z limits vertical dilution and also limits how rapidly an elevated plume mixes to the surface.

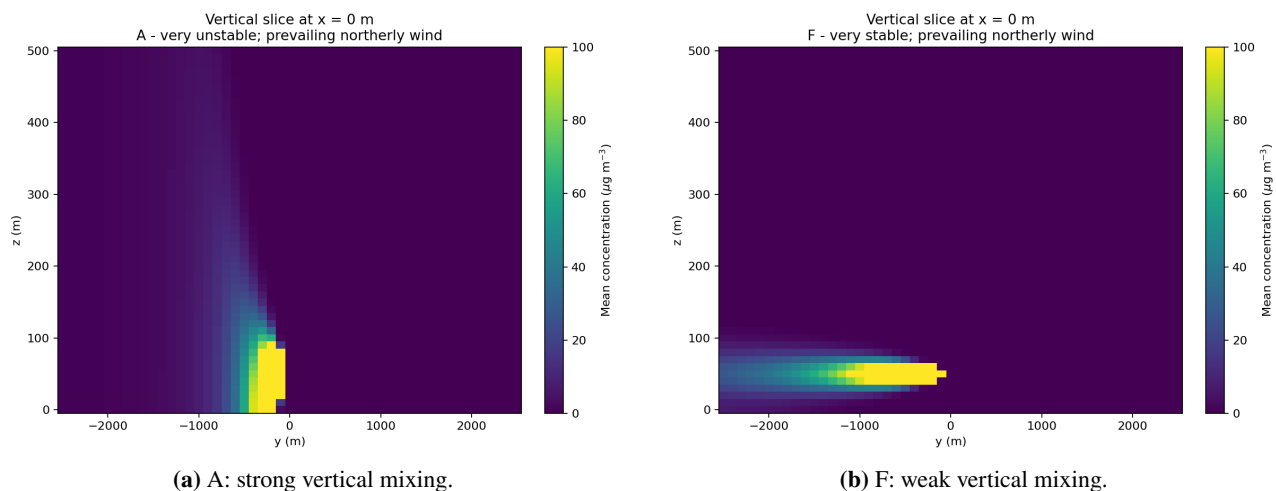


Figure 4: Vertical slices show directly how σ_z changes with stability.

1.5 An idealised annual cycle in stability

Exercise 5. Describe the seasonal pattern at the receptor and explain why instability and receptor concentration are not monotonically related. What is realistic and what is artificial about the experiment?

Model answer. The receptor shows its largest concentrations mainly during the more stable winter parts of the imposed cycle, with much lower 24-hour mean concentrations through the strongly unstable summer period. At this fixed point, concentration depends on whether the plume reaches the receptor as well as on how strongly it is diluted. A narrow stable plume can give a high concentration when its centreline passes over the receptor, whereas a very unstable plume can be mixed over such a large horizontal and vertical volume that the concentration at that one point is smaller. The realistic element is the broad seasonal tendency for stability to differ between winter and summer. The deliberately artificial elements are the simple prescribed sequence of stability classes, the lack of a diurnal cycle, fixed emissions, and the idealised meteorology rather than observed year-long weather.

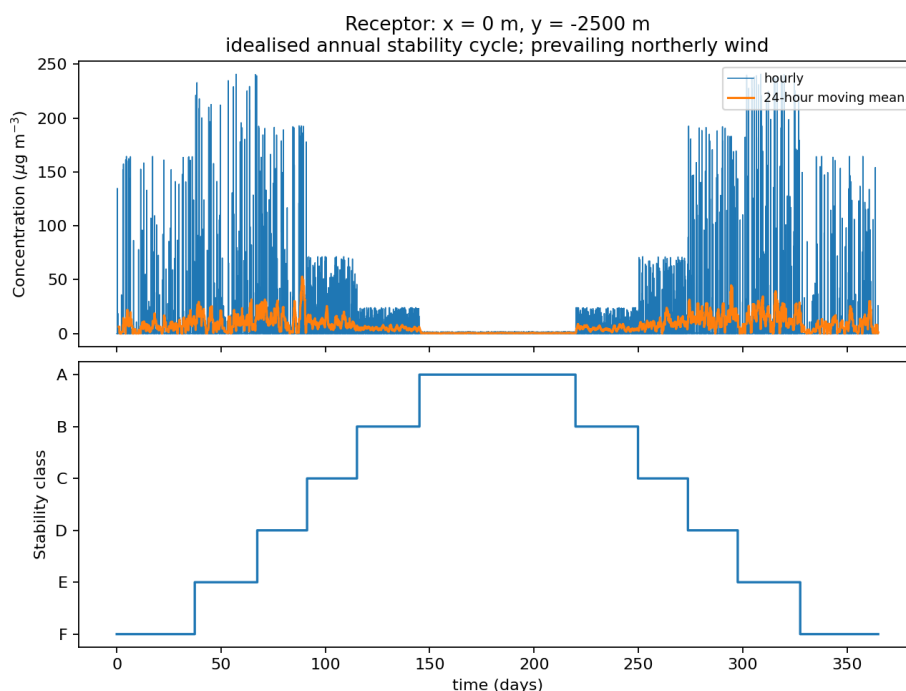


Figure 5: Idealised annual stability cycle and concentration at one receptor.

1.6 Idealised aerosol water uptake and map overlay

Prediction. Calculate the wet/dry mass factor for sodium chloride at $RH = 0.90$ using $\nu = 2$ and $M_s = 58.44 \text{ g mol}^{-1}$.

Expected prediction. Using

$$G_m = 1 + \frac{RH}{1 - RH} \nu \frac{M_w}{M_s},$$

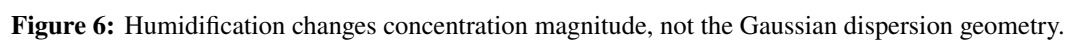
with $M_w = 18.015 \text{ g mol}^{-1}$ gives

$$G_m = 1 + 9 \times 2 \times \frac{18.015}{58.44} \approx 6.55.$$

The humidified mass concentration should therefore be about 6.55 times the dry mass concentration at every grid point. The spatial dispersion pattern should not change; only the concentration magnitude should be multiplied. With fixed contour values, the wet contours consequently extend farther from the source.

Exercise 6. Does water uptake change the dispersion pattern itself?

Model answer. No. In this teaching model G_m is applied as a multiplicative mass factor after the Gaussian dispersion calculation. The wind, σ_y , σ_z and source geometry are unchanged, so the normalised spatial pattern is identical. The wet map has larger concentrations everywhere by the same factor, causing any fixed-value contour to move outward.



Model answer. For $\nu = 1$ and $M_s = 200 \text{ g mol}^{-1}$ at RH = 0.90,

The organic-acid wet/dry mass enhancement is therefore much smaller than the sodium-chloride value of about 6.55 under the idealised solution model.